

Pune District Digital Soil Mapping and Gat-wise Digital Soil Health Card System

Comprehensive Technical Project Proposal & Operational Implementation Framework

Platform: Google Earth Engine (GEE) + Python | Sampling: Conditioned Latin Hypercube (cLHS) | Spatial Resolution: Cadastral Gat/Parcel Level

Executive Summary

This technical report presents a scientifically structured framework for developing a district-scale Digital Soil Mapping (DSM) and Gat-wise Digital Soil Health Card system for Pune District using an integrated Google Earth Engine (GEE) and Python workflow. The framework combines quality-controlled laboratory soil observations, systematically designed additional field sampling, satellite observations, terrain, climate, land-cover and parent-material information, followed by spatially robust machine-learning modelling and uncertainty assessment. The methodology is explicitly grounded in the SCORPAN framework and incorporates lessons from the systematic review of Indian DSM by Dash et al. (2022).

A central component is representative and adaptive sampling rather than a fixed samples-per-hectare rule. Existing observations will first be audited for coordinate quality, depth consistency, analytical methods, units, duplication, outliers and spatial clustering. Environmental representativeness will then be assessed, and cLHS or partitioned/stratified cLHS will be used where justified. Bhattacharyya distance and/or Jensen-Shannon divergence will quantify feature-space representation, while learning curves based on R^2 , RMSE and MAE will support the practical sample-size decision. This directly addresses the Indian DSM gaps identified by Dash et al. (2022), including poor sampling density, limited independent validation, limited comparative modelling and insufficient uncertainty mapping.

GEE will be used for satellite, terrain and climate covariate preparation and extraction at soil locations. Python will be used for cLHS, feature-space analysis, sample-size optimization, machine-learning comparison, spatial validation and uncertainty analysis. Final validated predictions will be generated at the selected analysis resolution and intersected with authoritative Gat/parcel boundaries to provide transparent Gat-wise soil information, clearly distinguishing laboratory observations from model estimates.

Evidence Base and Methodological Rationale

The present revision incorporates the systematic review by Dash, Panigrahi and Mishra (2022), “Identifying opportunities to improve digital soil mapping in India”, published in *Geoderma Regional*. The review screened 35 peer-reviewed SCORPAN-based DSM studies in India and identified recurring gaps that are directly relevant to the proposed Pune District implementation.

Key evidence incorporated into this report:

Identified Research Gap in India (Dash et al., 2022)	Current Indian DSM Status	Proposed Pune District Technical Solution
Spatial Resolution & Landholding Scale	Most studies operate at coarse/regional scales (>250m - 1000m), ineffective for 1.08 ha smallholders.	High-resolution Sentinel-2 (10m) + DEM derivatives (10m-30m) integrated directly with cadastral Gat polygons.
Sampling Methodology	Utilizes random (58%), stratified random (31%), and grid (11%) sampling, often lacking a framework for adequate density.	Uses environmental representativeness and spatially robust validation rather than a single fixed sampling-density rule.
Mapping Depth	80% of studies focus only on surface soil, while just 20% map multiple depths.	Employs depth-standardized modelling with harmonized depth intervals and explicit depth metadata.
Environmental Covariate Scope	Rely heavily on Organism (O) and Relief (R) (29% use only these), neglecting Soil (S), Climate (C), and Parent Material (P).	Utilizes a broad, screened SCORPAN stack (multi-temporal Sentinel-2 indices, CHIRPS/ERA5 climate, GSI lithology, and legacy soil layers).
Target Variables	Focuses predominantly on soil carbon, depth, separates, macro-nutrients, pH, and EC; categorical soil classes are less explored.	Incorporates the GlobalSoilMap reference set (effective depth, sand, silt, clay, coarse fragments, effective CEC, bulk density, AWC, etc.).
Legacy Data Integration	Legacy data is valuable but often suffers from unstandardized sampling/analytical techniques and non-static temporal mismatches.	Treats legacy data as evidence requiring strict Quality Control (QC) and harmonization, not automatically equivalent to new observations.
Algorithm Evaluation & Comparison	Regression kriging is most frequent, but 86% of studies rely on a single ML model without benchmark testing.	Systematic multi-model benchmarking comparing various models (e.g., Random Forest, Support Vector Regression, XGBoost).
Validation Methodology	76% use simple random data splitting (causing spatial autocorrelation bias); only a small proportion (5%) use independent validation.	Prioritizes Spatial Block Cross-Validation and/or a 1000-point spatially independent hold-out test set.
Uncertainty Quantification	66% of Indian DSM articles provide no spatial uncertainty or confidence maps.	Deploys Quantile Regression Forests (QRF) to generate 90% prediction intervals and spatial confidence maps.
Practical Extension Utility	Maps remain in academic literature or non-downloadable formats without direct farmer linkage.	Gat-wise web/mobile retrieval interface: Farmer enters Village + Gat No. to get an instant calibrated Soil Health Card & STCR dosage.

1. Project Vision

To establish a scientifically validated Pune District Digital Soil Intelligence Platform that converts laboratory soil observations, satellite data and environmental information into spatially explicit, Gat-wise digital soil information accessible to farmers and agricultural institutions.

2. Main Goal

To develop Digital Soil Maps for priority soil-health parameters across Pune District and create a digital service in which an authorized user can retrieve Gat-wise soil information by selecting a village and Gat number.

3. Specific Objectives

1. Ingest, clean, and harmonize all existing village-level soil sample datasets into a georeferenced Master Soil Database.
2. Develop a cloud-based environmental covariate extraction pipeline in GEE covering all SCORPAN factors for Pune District.
3. Use cLHS or partitioned/stratified cLHS where justified by environmental heterogeneity.
4. Determine the mathematical optimum sample size using Bhattacharyya Distance (Khan et al., 2023) and Jensen-Shannon Divergence (Saurette et al., 2024).
5. Begin operational implementation in Baramati Tehsil and validate the complete end-to-end workflow before district-wide expansion.
6. Train and benchmark multiple ML models (RF, SVR, XGBoost) and quantify spatial uncertainty using Quantile Regression Forests.
7. Spatially intersect DSM raster outputs with cadastral Gat shapefiles in PostGIS to generate parcel-level statistics and automated Soil Health Cards.

4. Detailed Sampling Methodology: cLHS and PcLHS

Conditioned Latin Hypercube Sampling (cLHS) is an advanced stratified random sampling strategy that draws sample locations from multivariate environmental covariates such that the marginal distribution of each covariate and the multivariate correlation structure are maximally preserved (Minasny and McBratney, 2006; Moharana et al., 2024).

4.1 Step-by-Step cLHS / PcLHS Implementation

Phase 1: Project Initiation & Data Collation

- **Step 1: Project Setup & Boundary**
 - Establish the boundaries for Pune District, including taluka, village, and Gat/parcel databases.
- **Step 2: Existing Soil Data**
 - Compile legacy laboratory samples along with their coordinates, depth information, and analytical methods, followed by strict quality control (QC).

Phase 2: Feature Engineering & Pre-processing

- **Step 3: GEE Environmental Covariates (Feature Stack Construction)**
 - Extract and align multi-temporal Sentinel-2 imagery (NDVI, EVI, NDMI, SAVI) and high-resolution DEM derivatives (Elevation, Slope, Aspect, TWI, TPI, TRI, MRVBF, MRRTF).
 - Integrate bioclimatic grids, land use/land cover (LULC), and geology data to export a normalized raster stack.
- **Step 4: Covariate Screening (Multicollinearity)**
 - Calculate the Variance Inflation Factor (VIF) and correlation matrices in Python.
 - Systematically eliminate collinear variables with a VIF > 10 to avoid feature redundancy and model overfitting.

Phase 3: Advanced Sampling Strategy (cLHS / PcLHS)

- **Step 5: Representativeness Analysis (Evaluating Legacy Coverage)**
 - Project existing Pune soil sample locations into the complete multivariate environmental feature space.
 - Identify physiographic and vegetative zones that are currently under-represented.
- **Step 6: Stratification / PcLHS Decision (Regionalization)**
 - Address Pune District's spatial heterogeneity (e.g., Western Ghats, transition zone, and eastern plains) by applying REDCAP (Regionalization with Dynamically Constrained Agglomerative Clustering and Partitioning).
 - Partition the district into continuous homogeneous clusters to define zones and reduce prediction errors compared to global sampling.
- **Step 7: cLHS Sample-Size Optimization**
 - Run iterative candidate designs for sample sizes ($n = 50$ to 1500).
 - Calculate divergence metrics, specifically Bhattacharyya Distance and Jensen-Shannon Divergence (D_{JS}), to perform plateau analysis.
 - Identify the unit-invariant knee/elbow point where the cumulative distribution function reaches $\geq 95\%$ similarity with the district population.
- **Step 8: cLHS/PcLHS Point Selection**
 - Execute annealing optimization in Python (clhs-py) for 5000 iterations to allocate precise spatial coordinates for additional sampling.
 - Apply operational accessibility constraints, such as buffers along road networks and field access routes.
- **Step 9: Field & Laboratory Sampling**
 - Use a standardized Standard Operating Procedure (SOP) to collect 0-15 cm composite soil samples at the generated coordinates via GPS.
 - Analyze the collected samples for 12 essential parameters using standard wet chemistry and spectroscopic protocols.

Phase 4: Modeling & Delivery

- **Step 10: Python Machine Learning**
 - Train spatial prediction models using algorithms such as Random Forest (RF), Support Vector Regression (SVR), and Gradient Boosting.

- Implement spatial validation to assess model accuracy.
- **Step 11: Digital Soil Maps**
 - Generate spatial predictions and quantify uncertainty.
 - Conduct extrapolation checks to ensure model reliability across the boundary.
- **Step 12: Gat-wise Digital Soil Health Card**
 - Extract parcel statistics to generate a digital report for individual landholdings.
 - Include confidence intervals within the report to support farmer decision-making.

4.2 Complete Methodology: Graphical Representation

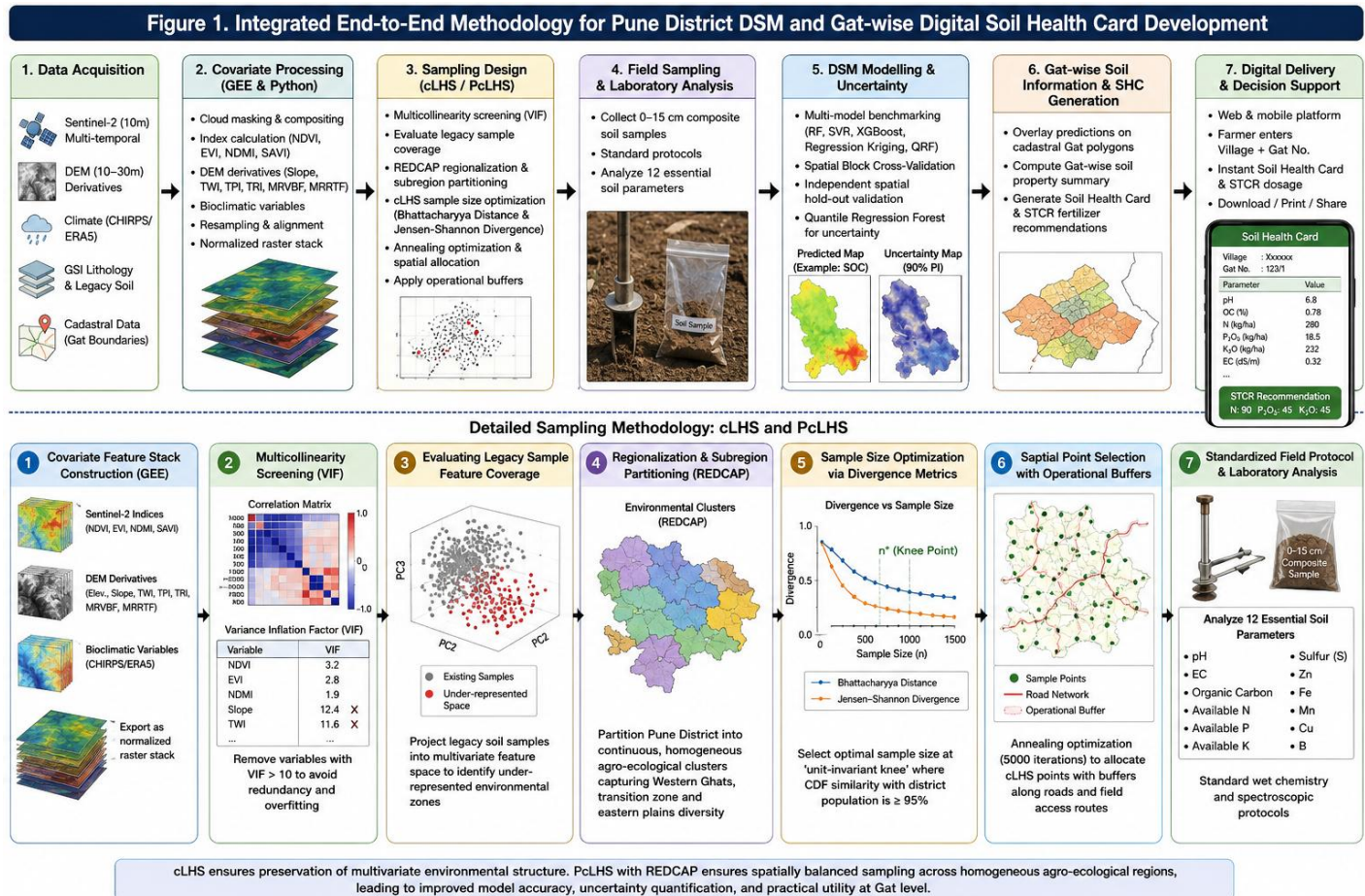


Figure 1. Integrated end-to-end methodology for Pune District DSM and Gat-wise Digital Soil Health Card development.

5. Scientific Framework: SCORPAN

Digital Soil Mapping predicts the spatial distribution of soil properties by establishing relationships between measured soil observations and environmental variables. The conceptual basis is commonly described through the SCORPAN framework: soil information (S), climate (C), organisms/vegetation (O), relief (R), parent material (P), age/time (A) and spatial position (N) (McBratney et al., 2003).

Factor	Pune DSM examples
S – Soil	Legacy laboratory observations and validated soil information
C – Climate	Rainfall and temperature summaries
O – Organisms	Vegetation indices and land-cover information
R – Relief	Elevation, slope, aspect and terrain indices
P – Parent material	Geology and lithology
A – Age	Time-related information where relevant
N – Spatial position	Spatial structure used cautiously

6. Integrated GEE + Python Architecture

Component	Platform	Main role
Satellite processing	GEE	Sentinel/Landsat composites and indices
Terrain and climate	GEE	Environmental covariates
Covariate extraction	GEE	Extract values at soil points
cLHS	Python	Representative sample selection
Sample-size optimization	Python	Distribution metrics and learning curves
ML modeling	Python	RF, SVR and boosting comparison
Spatial validation	Python	Spatial CV and independent testing
DSM maps	GEE/Python/GIS	Prediction and uncertainty
Gat reporting	GIS/PostGIS/Web	Parcel-level statistics and reports

7. Data Required

Category	Required data
Administrative	Pune district, taluka and village boundaries
Gat/parcel	Authoritative Gat polygons and unique Gat numbers
Soil observations	Coordinates, depth, date, laboratory values, methods and units
Satellite	Sentinel-2/Landsat time series
Terrain	DEM, elevation, slope, aspect and derivatives
Climate	Rainfall and temperature
LULC	Land-use/land-cover
Parent material	Geology/lithology
External soil products	SoilGrids and suitable local/open soil datasets

8. Detailed Procedure in Google Earth Engine

8.1 Study Area

Import and verify the Pune District boundary. Prepare taluka/village boundaries for reporting. Keep Gat boundaries as a dedicated parcel layer for the final reporting stage.

- Verify Pune District, taluka, village and Baramati administrative boundaries.
- Prepare Sentinel-2 Surface Reflectance imagery with cloud/cloud-shadow masking and scientifically justified seasonal/annual composites.

- Calculate relevant vegetation/moisture indices such as NDVI, EVI, NDMI and SAVI; retain only useful, non-redundant predictors.
- Prepare DEM-derived elevation, slope, aspect and other terrain variables where justified.
- Prepare climate summaries and document temporal period and spatial resolution.
- Integrate LULC and parent-material/geology layers where available.
- Align extent, projection and analysis resolution and maintain layer metadata.
- Extract covariates at cleaned soil sample locations and export the modelling table to Python.

9. Existing Soil Data Quality Control

- Assign a permanent unique Sample_ID.
- Validate coordinates and record coordinate reference system and positional quality where available.
- Standardize or harmonize sampling depths while retaining original depth information.
- Standardize units and document laboratory analytical methods.
- Record sample date, source, village, taluka, Gat number and land-use/crop information where available.
- Review duplicates, outliers and implausible observations using documented rules.
- Assess spatial clustering and identify areas with insufficient representation.
- Maintain QA/QC and chain-of-custody records for all new field observations.

10. Sampling Design and Sample-Size Decision

10.1 Representativeness Assessment

Existing observations will first be projected into the selected environmental feature space. Their marginal and multivariate distributions will be compared with the candidate population to identify environmental and geographic gaps. This step determines whether additional sampling is necessary and where it is most valuable.

10.2 Heterogeneity and Partitioning

Pune District contains substantial environmental variation. Partitioned cLHS/PcLHS will therefore be used only if analysis demonstrates that separate environmental populations or zones are justified. Regionalization may use physiography, agro-environmental zones, terrain/landscape clustering or other documented criteria.

10.3 Sample-Size Optimization

Candidate additional sample sizes will be evaluated using repeated sampling designs. Bhattacharyya distance and/or Jensen–Shannon divergence will assess representation, while model learning curves using R^2 , RMSE and MAE will assess whether additional observations materially improve prediction. The selected sample size will balance representation, predictive stability, uncertainty, budget and field feasibility.

No universal number such as one sample per hectare will be adopted as the scientific basis for Pune District. The Baramati pilot will generate an evidence-based sampling requirement for Baramati; the requirement will then be reviewed for each expansion unit.

10.4 cLHS/PcLHS Point Selection

After the sampling requirement is approved, cLHS/PcLHS will select locations representing the chosen environmental covariates. Optimization settings and random seeds will be documented for reproducibility. Accessibility constraints will be applied after environmental selection, and inaccessible locations will be replaced using nearby environmental feature-space equivalents rather than arbitrary relocation.

11. Field and Laboratory Protocol

- Use the project-approved standardized sampling SOP and target depth interval.
- Record actual GPS coordinates at the sampling location.
- Record village, taluka, Gat number, land use/crop, irrigation status where relevant, date, depth and field observations.
- Use permanent Sample_ID and complete chain-of-custody documentation.
- If a selected point is inaccessible, document the replacement and its environmental equivalence.
- Analyze core soil-health parameters using one approved project SOP or formally harmonized laboratory methods.
- Record analytical method, units, laboratory, analyst/date, detection/reporting limits where applicable and QA/QC information.
- Perform duplicate and outlier review before observations enter the DSM training database.

12. Priority Soil Parameters

Group	Parameters
Core	pH, EC, Organic Carbon/SOC
Primary nutrients	Available N, P and K
Secondary	S, Ca and Mg where reliable
Micronutrients	Zn, Fe, Mn, Cu and B where required
Physical	Sand, silt, clay, texture and bulk density where available
Advanced	CEC and other indicators where sufficient observations exist

The exact parameter set for the Baramati pilot should be frozen before sampling. Only parameters with adequate, quality-controlled observations and suitable predictor relationships should be taken forward to predictive mapping.

13. Python Machine Learning and Statistical Analysis

The training table will combine laboratory-tested soil observations with environmental covariates extracted from GEE. Each soil property should normally be modeled separately.

Model	Role
Random Forest Regressor	Primary nonlinear DSM benchmark

SVR	Nonlinear comparison; requires appropriate scaling
Gradient Boosting/XGBoost	Strong comparative predictive model
Simple baseline	Optional benchmark

Model selection should be based on spatially robust performance rather than deciding in advance that one algorithm is universally best.

Each soil property will normally be modelled separately. Hyperparameters will be tuned using training data only. Model selection will be based on spatially robust performance rather than an assumed universally best algorithm.

- Spatial block cross-validation to reduce spatial autocorrelation bias.
- Spatially separated independent hold-out data where feasible.
- Metrics: R^2 , RMSE, MAE and Bias/Mean Error.
- Residual diagnostics and examination of model failure areas.
- Prediction uncertainty/confidence estimation where implemented.
- Extrapolation/domain checks to identify areas outside the training feature space.

14. SoilGrids and Other External Soil Data

SoilGrids and similar products are modelled estimates and will not be treated as ground truth. Pune laboratory observations remain the primary ground evidence, followed by validated new field observations. External products may be used for benchmarking, pattern comparison or as ancillary predictors only after independent testing and appropriate harmonization.

15. Digital Soil Map Production

- Select the best validated model separately for each soil property.
- Prepare the final predictor stack for the relevant implementation unit.
- Generate predictions over the approved analysis grid.
- Generate uncertainty/confidence layers.
- Perform extrapolation checks and flag low-support areas.
- Document depth, units, time period, resolution, predictor version and model version.
- Retain a clear distinction between observed soil values and model predictions.

16. Gat-wise Digital Soil Health Card

Authoritative Gat polygons will be overlaid with validated DSM layers. Because a Gat can contain multiple prediction cells, parcel-level statistics such as median/mean values and uncertainty measures will be calculated. Gat-level reporting is a reporting and decision-support layer; it does not imply that the soil property was directly measured at every location within the Gat.

- Village and Gat number

- Gat area and location
- pH, EC, SOC/OC and mapped nutrient estimates
- Status category using approved soil-health thresholds
- Within-Gat variability where useful
- Prediction confidence/uncertainty
- Model version and date
- Explicit label identifying laboratory-observed versus model-estimated values

User flow: Select Village → Enter Gat Number → Retrieve Gat → Overlay DSM layers → Calculate Gat statistics → Generate Digital Soil Health Card.

17. Technology Architecture

Component	Platform	Main Role
Environmental processing	Google Earth Engine	Satellite, terrain, climate and covariate preparation
Sampling & statistics	Python	cLHS/PcLHS, representativeness, sample-size analysis and learning curves
Machine learning	Python	RF, SVR, boosting/XGBoost and validation
Spatial database	PostgreSQL/PostGIS	Gat geometry, soil observations, predictions and parcel statistics
GIS/map services	GIS/Web services	Visualization and map delivery
Application layer	API + Web/Mobile	Village/Gat query and Soil Health Card retrieval
Reproducibility	Version control + metadata	Dataset, code, model and report versioning

18. Implementation Timeline – Pune District with Baramati as the First Operational Unit

The original single 12-month district timeline has been revised because the project will not attempt full Pune field implementation at once. Pune-wide framework and data preparation will start first, while Baramati will be the first complete pilot. After Baramati validation, the workflow will be scaled to additional talukas and then consolidated at district level.

Phase / Period	Geographic Scope	Major Activities	Key Deliverables
Phase 0 – Month 1	Pune District + Baramati	Project inception; administrative/Gat data inventory; existing soil-data inventory; SOP and metadata finalization; database design; identify Baramati data sources.	Approved project framework, data inventory, SOP and database schema

Phase 1 – Months 1–2	Baramati	Baramati soil-data QC; coordinate/depth/unit/method harmonization; GEE covariate stack; preliminary representativeness and spatial-gap analysis.	Baramati QC soil database, covariate stack and gap assessment
Phase 2 – Months 2–3	Baramati	Heterogeneity assessment; cLHS/PcLHS decision; candidate sample-size evaluation using distribution metrics and learning curves; final sampling design.	Approved Baramati sample requirement and sampling locations
Phase 3 – Months 3–5	Baramati	Field sampling, GPS capture, laboratory analysis, QA/QC and database integration.	Validated Baramati field/lab dataset
Phase 4 – Months 5–7	Baramati	Property-wise RF/SVR/XGBoost modelling; spatial cross-validation; independent validation where feasible; uncertainty and extrapolation assessment.	Validated Baramati models and performance report
Phase 5 – Months 7–8	Baramati	Generate DSM layers; Gat overlay; parcel statistics; Digital Soil Health Card prototype; user testing.	Baramati DSM + Gat-wise Soil Health Card pilot
Phase 6 – Months 8–9	Pune District / selected next talukas	Review Baramati lessons; update SOP/pipeline; prioritize next talukas; run legacy-data and representativeness checks for each selected unit.	Scale-up plan and updated sampling designs
Phase 7 – Months 9–11	Additional talukas	Sequential sampling, laboratory analysis, modelling and validation using the standardized pipeline; adaptive sampling where required.	Validated taluka DSM outputs
Phase 8 – Months 11–12+	Pune District	Harmonize taluka outputs; district-level validation and uncertainty review; integrate Gat-wise reporting; finalize platform and SOPs.	Pune District DSM framework, integrated reporting system and final implementation documentation

19. Detailed Baramati Pilot Work Plan

Week / Period	Activity	Decision / Output
Weeks 1–2	Collect all available Baramati soil records and Gat/village boundaries; establish Master Soil Database.	Complete inventory and data structure

Weeks 2–4	QC coordinates, depth, units, methods, duplicates/outliers and spatial clustering.	QC-approved legacy dataset
Weeks 2–5	Prepare Baramati satellite, terrain, climate, LULC and parent-material covariates in GEE.	Initial covariate stack
Weeks 4–6	Assess feature-space representativeness and spatial gaps; evaluate heterogeneity.	Sampling strategy decision
Weeks 5–7	Run candidate sample-size designs; compare distribution metrics and learning curves.	Evidence-based sample-size recommendation
Weeks 7–8	Run cLHS/PcLHS and accessibility checks; finalize field points.	Approved sampling map/list
Weeks 9–16	Field collection, laboratory analysis, QA/QC and data integration.	New validated observations
Weeks 17–24	Train/tune candidate models and conduct spatial validation.	Model comparison and selected model
Weeks 23–28	Generate DSM and uncertainty maps; conduct extrapolation checks.	Validated Baramati DSM
Weeks 27–32	Gat overlay, parcel statistics and Digital Soil Health Card prototype.	Operational Baramati pilot
Weeks 31–34	Technical review, user testing and documentation of lessons for scale-up.	Baramati pilot completion and Pune scale-up SOP

20. Expansion Strategy After Baramati

- Use Baramati to validate the complete workflow, software pipeline, database structure, field SOP and reporting format.
- For every new taluka, repeat only the necessary data-audit and representativeness checks.
- Reuse common Pune-wide covariate definitions, metadata standards, code modules and validation protocols.
- Adapt sampling intensity when the new taluka has different environmental heterogeneity or legacy-data coverage.
- Maintain a common model/data versioning system so taluka outputs remain traceable.
- Integrate taluka-level outputs into the district platform only after quality and validation checks.

21. Research Gaps Addressed

- Standardized and traceable soil database development.
- Empirical sampling-density determination instead of fixed density assumptions.
- Integration of relevant SCORPAN environmental covariates.
- Comparative machine-learning evaluation.
- Spatially robust validation and independent testing where feasible.
- Prediction uncertainty and extrapolation assessment.
- Gat-wise translation of validated predictions into an end-user service.

22. Expected Outputs

Output	Description
Master Soil Database	QC and harmonized Pune/implementation-unit soil observations with metadata
Environmental Covariate Stack	Documented GEE predictor layers
Sampling Design	Evidence-based cLHS/PcLHS locations and sample-size rationale
Laboratory Dataset	Validated new observations with QA/QC
Model Performance Report	Spatial validation metrics, residual diagnostics and model selection
Digital Soil Maps	Validated soil-property prediction layers with uncertainty
Gat-wise Soil Health Cards	Parcel-level soil estimates with confidence and observed/modelled labels
Web/Mobile Prototype	Village + Gat based retrieval interface
SOP & Scale-up Framework	Reusable process for expansion from Baramati to Pune District

26. References

McBratney, A.B., Mendonça Santos, M.L. and Minasny, B. (2003). On digital soil mapping. *Geoderma*, 117, 3–52.

Minasny, B. and McBratney, A.B. (2006). A conditioned Latin hypercube method for sampling in the presence of ancillary information. *Computers & Geosciences*, 32, 1378–1388.

Khan, A., Aitkenhead, M., Stark, C.R. and Jorat, M.E. (2023). Optimal sampling using Conditioned Latin Hypercube for digital soil mapping: An approach using Bhattacharyya distance. *Geoderma*, 439, 116660.

Huang, B., Yang, G., Lei, J. and Wang, X. (2025). A partitioned conditioned Latin hypercube sampling method considering spatial heterogeneity in digital soil mapping. *Scientific Reports*, 15, 12851.

Saurette, D.D., Heck, R.J., Gillespie, A.W., Berg, A.A. and Biswas, A. (2024). Sample Size Optimization for Digital Soil Mapping: An Empirical Example. *Land*, 13, 365.

Moharana, P.C., Yadav, B., Meena, R.L., Nogiya, M., Malav, L.C., Tailor, B.L., Biswas, H. and Patil, N.G. (2024). Sampling Strategy for Digital Soil Mapping in the Thar Desert Region of India: A Conditioned Latin Hypercube Sampling Approach. *Journal of the Indian Society of Soil Science*, 72(2), 133–144.

Dash, P.K., Panigrahi, N. and Mishra, A. (2022). Identifying opportunities to improve digital soil mapping in India: A systematic review. *Geoderma Regional*, 28, e00478. <https://doi.org/10.1016/j.geodrs.2021.e00478>.